multiplication. Examples of systems that possess such symmetry and the corresponding closed *Gaussian surfaces* we will use to surround them are summarized below:

Symmetry	System	Gaussian Surface
Cylindrical	Infinite line	Coaxial Cylinder
Planar	Infinite plane	Gaussian "Pillbox"
Spherical	Sphere, Spherical shell	Concentric Sphere

Solving Problems using Gauss's law

Gauss's law provides a powerful tool for calculating the electric field of charge distributions that have one of the three symmetries listed above. The following steps are useful when applying Gauss's law:

(1) Identify the symmetry associated with the charge distribution, and the associated shape of "Gaussian surfaces" to be used.
(2) Divide space into different regions associated with the charge distribution, and determine the exact Gaussian surface to be used for each region. The electric field must be constant or known (i.e. zero) across the Gaussian surface.
(3) For each region, calculate q_{enc}, the charge enclosed by the Gaussian surface.
(4) For each region, calculate the electric flux Φ_E through the Gaussian surface.
(5) Equate Φ_E with $q_{\text{enc}}/\varepsilon_0$, and solve for the electric field in each region.

Important Equations

Electric flux through a surface *S*: $\Phi_E = \iint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$

Gauss's law: $\Phi_E = \oiint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{enc}}}{\varepsilon_0}$

Important Concepts

Gauss's Law applies to closed surfaces—that is, a surface that has an inside and an outside (e.g. a basketball). We can compute the electric flux through any surface, open or closed, but to apply Gauss's Law we must be using a closed surface, so that we can tell how much charge is inside the surface.

Gauss's Law is our first Maxwell's equations, and concerns closed surfaces. Another of Maxwell's equations, the magnetic Gauss's Law, $\Phi_B = \oiint_S \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$, also applies to a closed surface. Our third and fourth Maxwell's equations will concern open surfaces, as we will see.